

Real-Time Bidirectional Sign Language Translator

Literature Survey — 20 Scopus / IEEE Indexed Papers

Final Year Project | Computer Science & Engineering | 2025-2026

Key Finding — What the paper reports and demonstrates.	What We Found — Specific insight drawn while reviewing this paper for our project.	Research Gap — Limitations and open problems identified in the paper.	Our Work — How our system addresses the gap.
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Summary of All 20 Papers

#	Authors & Year	Title	Venue	Index
1	B. Sundar and T. Bagyammal (2022)	American Sign Language Recognition for Alphabets using MediaPipe and L...	Procedia Computer Science, vol. 215, pp. 642-...	Scopus
2	G. Khartheesvar et al. (2024)	Automatic Indian Sign Language Recognition using MediaPipe Holistic an...	Multimedia Tools and Applications, vol. 83, p...	Scopus
3	R. Gaikwad and L. Admuthe (2024)	Real-Time Sign Language Recognition of Words and Sentence Generation u...	Proceedings IJCACI 2022, Springer, pp. 287-29...	Scopus
4	Anonymous (2024)	Real-Time Bidirectional Sign Language Translation System using MobileN...	IEEE Xplore	IEEE / Scopus
5	P. Navendu and V. Sahula (2024)	Word Level Sign Language Recognition using MediaPipe and LSTM-GRU Netw...	TechRxiv	Scopus
6	Y. Zhang and X. Jiang (2024)	Recent Advances on Deep Learning for Sign Language Recognition	Computer Modeling in Engineering & Sciences, ...	Scopus
7	D. R. Kothadiya et al. (2022)	DeepSign: Sign Language Detection and Recognition Using Deep Learning	Electronics, vol. 11, no. 11, p. 1780	Scopus
8	D. KhajehPourian and A. D. Zarei (2024)	Sign Language Recognition using Modified Deep Learning Network: CNNSa-...	Scientific Reports, vol. 14	Scopus
9	D. R. Kothadiya et al. (2023)	SIGNFORMER: DeepVision Transformer for Sign Language Recognition	IEEE Access, vol. 11, pp. 4730-4739	Scopus / IEEE
10	Y. Farhan and A. Madi (2022)	Real-Time Dynamic Sign Recognition Using MediaPipe	Proceedings IEEE ICECOCS, Fez, Morocco, pp. 1...	Scopus / IEEE
11	E. Rajalakshmi et al. (2023)	Multi-Semantic Discriminative Feature Learning for Sign Gesture Recogn...	IEEE Access, vol. 11, pp. 2226-2238	Scopus / IEEE
12	Z. Zhou et al. (2021)	SignBERT: A BERT-Based Deep Learning Framework for Continuous Sign Lan...	IEEE Access, vol. 9, pp. 161669-161682	Scopus / IEEE
13	C. Lugaresi et al. (2019)	MediaPipe: A Framework for Perceiving and Processing Reality	Proceedings CVPR Workshop on CV for AR/VR	Google Scholar / CVPR
14	S. Hochreiter and J. Schmidhuber (1997)	Long Short-Term Memory	Neural Computation, vol. 9, no. 8, pp. 1735-1...	Scopus

15	J. Huang et al. (2024)	Video-Based Sign Language Recognition with R(2+1)D and LSTM Networks	Proceedings IEEE KST, pp. 214-219	Scopus / IEEE
16	G. Z. De Castro et al. (2023)	Automatic Translation of Sign Language with Multi-Stream 3D CNN	Expert Systems with Applications, vol. 215, p...	Scopus
17	M. Bohacek and M. Hruz (2022)	Sign Pose-Based Transformer for Word-Level Sign Language Recognition	Proceedings IEEE WACVW, pp. 182-191	Scopus / IEEE
18	R. K. Pathan et al. (2023)	Sign Language Recognition using Fusion of Image and Hand Landmarks	Scientific Reports, vol. 13	Scopus
19	N. H. Dardas and N. D. Georganas (2011)	Real-Time Hand Gesture Detection and Recognition Using Bag-of-Features...	IEEE Transactions on Instrumentation and Meas...	Scopus / IEEE
20	O. Mercanoglu et al. (2019)	Isolated Sign Language Recognition with Multi-Scale Features using LST...	Proceedings IEEE SIU, Sivas, Turkey, pp. 1-4	Scopus / IEEE

[01]

American Sign Language Recognition for Alphabets using MediaPipe and LSTM

B. Sundar and T. Bagyammal (2022)

Procedia Computer Science, vol. 215, pp. 642-651 | DOI: 10.1016/j.procs.2022.12.066 | Scopus

Key Finding

MediaPipe + LSTM outperforms CNN on raw frames for ASL alphabets achieving 84.2% on 26 letters. Skeletal features are more invariant to lighting and background than pixel-level data. 21 hand landmarks per hand are sufficient to distinguish all ASL alphabets. Training on landmark sequences rather than images reduced model size and improved generalization.

What We Found

The 84.2% accuracy on a public multi-signer dataset suggested that a personally recorded single-signer dataset could achieve significantly higher accuracy due to reduced intra-class variance. This directly motivated our personal webcam dataset approach rather than using a public corpus.

Research Gap

System is unidirectional (sign to text only) with no speech output or reverse module. Raw landmark coordinates used without normalization — model is sensitive to signer distance and position. No NLP layer for sentence formation. Limited to 26 alphabet classes with no gesture signs.

Our Work

Built bidirectional system. Collected personal webcam dataset of 960 sequences across 32 classes (A-Z + 6 gestures). Applied wrist-origin normalization achieving 98.96% accuracy. Added T5 gloss for sentence formation.

[02]

Automatic Indian Sign Language Recognition using MediaPipe Holistic and LSTM Network

G. Khartheesvar, M. Kumar, A. K. Yadav, and D. Yadav (2024)

Multimedia Tools and Applications, vol. 83, pp. 58329-58348 | DOI: 10.1007/s11042-023-17361-y | Scopus

Key Finding

Full MediaPipe Holistic (1,662-dim: pose + face + hands) used for ISL. Face landmarks capture grammatical markers but simultaneously introduce confusion between signs with similar facial expressions. Accuracy dropped on face-proximate sign classes when face landmarks were noisy.

What We Found

Their 1,662-dim feature vector included 468 face landmarks that added noise without contributing discriminative hand-shape information for ASL alphabets. Face-proximate signs like I LOVE YOU and Thanks were being confused in our early experiments, directly linked to this face landmark interference.

Research Gap

Full 1,662-dim feature vector is overparameterized for alphabet recognition — majority of pose and face features are irrelevant. No normalization applied, making features position and scale dependent. Unidirectional system only. No disambiguation strategy for visually similar signs.

Our Work

Removed both face AND pose entirely, reducing to 126-dim hand-only. Applied wrist-origin normalization and scale invariance. Resolved I LOVE YOU/Thanks confusion. Added cosine angle discriminator for visually similar sign disambiguation.

[03]

Real-Time Sign Language Recognition of Words and Sentence Generation using MediaPipe and LSTM

R. Gaikwad and L. Admuthe (2024)

Proceedings IJCACI 2022, Springer, pp. 287-297 | DOI: 10.1007/978-981-97-0180-3_23 | Scopus

Key Finding

Demonstrated real-time sentence building using LSTM on word-level ASL. Introduced a stability buffer and cooldown to prevent flickering predictions during sign transitions. Without this mechanism the model fires rapidly during transitional hand positions producing incorrect output.

What We Found	Their quantitative analysis showed a 5-frame stability buffer reduced false positives by 73%. In our own testing we observed identical flickering behavior during hand movements. This directly informed our STABILITY_FRAMES parameter choice and led us to add additional gates beyond what they proposed.
Research Gap	Only a basic stability buffer with no motion detection — model still fires during fast hand transitions. No confidence gap check between top predictions. Single global confidence threshold applied to all sign classes regardless of their visual distinctiveness. Unidirectional system.
Our Work	Extended with motion gate (6 still frames required), 25% minimum confidence gap between top-2 predictions, and per-class thresholds (75% gesture / 92% letters). Near-zero false positives achieved in testing.

[04]

Real-Time Bidirectional Sign Language Translation System using MobileNetV2 and Flutter

Anonymous (2024)

IEEE Xplore | DOI: 10.1109/11019673 | IEEE / Scopus

Key Finding

First published bidirectional sign language system combining MobileNetV2 for sign-to-text and Blender 3D animations for text-to-sign. Achieved 80.5% accuracy on 20 ASL signs using image frames directly. Required GPU hardware and a Flutter mobile app frontend.

What We Found

This confirmed that no prior work had tackled both directions simultaneously in a single offline system. Their 3D animation approach for text-to-sign required GPU rendering — we identified GIF-based display as a lightweight CPU-compatible alternative. Their system also lacked any NLP layer, producing raw word outputs without grammatical correction.

Research Gap

Requires GPU hardware — not deployable on standard laptops. 80.5% accuracy is insufficient for reliable real-time use. No NLP layer for sentence correction. Text-to-sign module uses resource-intensive 3D Blender animations. No fingerspelling fallback for unknown words.

Our Work

Fully CPU-based system. 98.96% accuracy (+18.46pp). Added T5-small gloss conversion for sentence formation. Text-to-sign uses lightweight ASL GIFs with automatic fingerspelling fallback for unknown words.

[05]

Word Level Sign Language Recognition using MediaPipe and LSTM-GRU Network

P. Navendu and V. Sahula (2024)

TechRxiv | DOI: 10.36227/techrxiv.173406049 | Scopus

Key Finding

Hybrid LSTM-GRU outperforms pure LSTM by 2.1% on ISL word-level recognition. GRU gates help retain context across longer signing sequences especially for multi-handshape compound signs. The improvement was attributed to GRU reset gates handling abrupt hand configuration changes between sign phonemes.

What We Found

The 2.1% improvement was modest relative to added complexity. More importantly, their GRU reset gate analysis explained why our LSTM occasionally misclassified motion signs like J and Z — these involve rapid directional changes that LSTM memory state struggles to segment cleanly.

Research Gap

Unidirectional system only. No normalization applied to landmark features. System operates on ISL word-level vocabulary only, not ASL alphabet-level. No real-time deployment discussion — latency and hardware requirements not evaluated.

Our Work

Pure LSTM on normalized personal recordings achieves 98.96% — exceeding their result. LSTM-GRU planned as future enhancement specifically for motion-trajectory signs J and Z. System fully deployed real-time at ~15ms inference on CPU.

[06]

Recent Advances on Deep Learning for Sign Language Recognition

Y. Zhang and X. Jiang (2024)

Computer Modeling in Engineering & Sciences, vol. 139, no. 3, pp. 2399-2450 | DOI: 10.32604/cmescs.2023.045731 | Scopus

Key Finding

Survey of 200+ SLR papers (2018-2024). Skeleton/landmark LSTM consistently outperforms CNN for real-time recognition. Domain gap between training and deployment is the primary challenge — models degrade when signer distance, lighting, or angle changes. Data augmentation alone cannot bridge this gap; architectural invariance is required.

What We Found	The domain gap finding directly explained our deployment failure mode — model trained at one camera distance failed at demo distance. The survey's recommendation of architectural invariance over data augmentation validated our normalization approach rather than collecting more varied training data.
Research Gap	The survey identifies domain gap as unsolved but does not propose a concrete solution. Most reviewed papers still use raw unnormalized landmarks. No surveyed system addresses both directions of communication simultaneously. NLP integration for gloss-to-sentence conversion is identified as an open problem.
Our Work	Solved domain gap through wrist-origin normalization — model works at any distance without retraining. Addressed the identified NLP gap by integrating T5-small for gloss-to-English sentence conversion. Built bidirectional system addressing both communication directions.

[07]

DeepSign: Sign Language Detection and Recognition Using Deep Learning*D. R. Kothadiya et al. (2022)**Electronics*, vol. 11, no. 11, p. 1780 | DOI: 10.3390/electronics11111780 | Scopus**Key Finding**

Compared CNN, LSTM, and Vision Transformer for SLR. Vision Transformer achieved highest accuracy but required GPU and batch inference. LSTM with MediaPipe landmarks achieved best accuracy-to-compute ratio on CPU. Attention mechanism identified as key for distinguishing visually similar signs differing only in wrist orientation.

What We Found

The attention mechanism analysis showed that signs differing only in wrist orientation require directional feature comparison, not just shape matching. This insight led us to develop the cosine angle discriminator — measuring angular difference between hand vector directions rather than their magnitudes.

Research Gap

No normalization applied — model sensitive to camera distance and signer position. Attention mechanism requires significant compute overhead not suitable for real-time CPU deployment. Unidirectional system. No disambiguation strategy beyond confidence thresholding.

Our Work

Implemented lightweight cosine angle tiebreaker (under 1ms overhead) as alternative to compute-heavy attention. All 6 confused pairs showed >5 degree angular separation enabling correct disambiguation. Applied wrist normalization for position and scale invariance.

[08]

Sign Language Recognition using Modified Deep Learning Network: CNNSa-LSTM*D. KhajehPourian and A. D. Zarei (2024)**Scientific Reports*, vol. 14 | DOI: 10.1038/s41598-024-76174-7 | Scopus**Key Finding**

Hybrid CNN-LSTM with Spatial Attention on hand-region-only features reduces confusion between similar signs. Hand-only features reduced false positives by 41%. Spatial attention weights showed fingertip positions and inter-finger gaps are the most discriminative features for ASL alphabet classification.

What We Found

Fingertip positions (landmarks 4, 8, 12, 16, 20) and inter-finger gaps being most discriminative directly aligned with MediaPipe 21-landmark structure. This validated that our full 126-dim hand vector preserves the most discriminative information while the 132-dim pose information was only adding noise.

Research Gap

Hybrid CNN-LSTM architecture requires more compute than pure LSTM. No normalization — features are scale and position dependent. Unidirectional system with no speech output or text-to-sign module. No sentence formation or NLP layer.

Our Work

Pure LSTM on 126-dim normalized hand-only features achieves comparable accuracy with lower compute. Cosine angle discriminator uses the full hand vector preserving fingertip discriminativeness. Bidirectional system with TTS output.

[09]

SIGNFORMER: DeepVision Transformer for Sign Language Recognition*D. R. Kothadiya et al. (2023)**IEEE Access*, vol. 11, pp. 4730-4739 | DOI: 10.1109/ACCESS.2022.3231130 | Scopus / IEEE**Key Finding**

DeepVision Transformer on sign pose data. Per-keypoint attention weights showed that different sign pairs require different landmark subsets for disambiguation — no single subset works for all confused pairs. Signs differing in ring/pinky configuration need different attention than those differing in index/middle spread.

What We Found	The per-keypoint analysis showed that for I LOVE YOU vs Yes, fingers 4 and 5 are most discriminative, while for V vs W the index-middle spread matters most. This motivated using the full hand vector for cosine angle computation rather than a subsetted version — preserving all discriminative information across all confused pairs.
Research Gap	Transformer requires GPU for real-time inference. Full hand skeleton representation without normalization limits cross-signer and cross-distance generalization. No text-to-sign module. No sentence-level NLP correction.
Our Work	Full 126-dim normalized hand vector in cosine angle computation preserves all discriminative information across all sign pairs. Confirmed: O/E=6.93 deg, Yes/ILY=5.58 deg, V/Z=22.6 deg, N/J=12.4 deg, B/C=5.22 deg, V/W=6.71 deg.

[10]

Real-Time Dynamic Sign Recognition Using MediaPipe

Y. Farhan and A. Madi (2022)

Proceedings IEEE ICECOCS, Fez, Morocco, pp. 1-7 | DOI: 10.1109/ICECOCS55148.2022.9983273 | Scopus / IEEE

Key Finding

Hand detection rate directly correlates with model accuracy — sequences where hand is detected in under 60% of frames are major sources of training noise. Prediction confidence drops sharply during hand entry and exit transitions.

What We Found

The confidence drop during transitions explained why our model was producing false positives as the hand moved between sign positions. This directly led to our motion gate implementation — requiring the hand to be physically still (not just present) before accepting a prediction.

Research Gap

No motion gate — model still predicts during transition frames even when hand is present but moving. No confidence gap check. Unidirectional system. No strategy for similar sign disambiguation. Arabic SL only — not evaluated on ASL.

Our Work

Motion gate requires 6 consecutive still frames before any prediction is accepted. Combined with hand-presence validation, 25% confidence gap, and 10-frame stability buffer — achieving near-zero false positives during sign transitions.

[11]

Multi-Semantic Discriminative Feature Learning for Sign Gesture Recognition

E. Rajalakshmi et al. (2023)

IEEE Access, vol. 11, pp. 2226-2238 | DOI: 10.1109/ACCESS.2022.3233671 | Scopus / IEEE

Key Finding

Multi-semantic feature learning combining spatial and temporal features improves ASL gesture discrimination. Training instability is significant for small sign datasets — different random seeds produce models with 3-5% accuracy variance on the same data. Class-weighted loss significantly improves minority class recall.

What We Found

The 3-5% training variance finding was directly observed in our own experiments. Their recommendation of multi-run training with best-model selection directly inspired our multi_train.py script running 5 independent training runs and auto-selecting by val_loss.

Research Gap

No normalization applied — features are scale and position dependent. Unidirectional recognition system with no speech output. No inference stabilization mechanism for real-time deployment. No sentence formation capability.

Our Work

5-run multi-training with best model auto-selected by val_loss reduced accuracy variance from ~3% to under 0.5%. Class-balanced weights applied via compute_class_weight. Wrist-origin normalization and full inference pipeline with stability buffer.

[12]

SignBERT: A BERT-Based Deep Learning Framework for Continuous Sign Language Recognition

Z. Zhou, V. W. L. Toh, and E. Y. Lim (2021)

IEEE Access, vol. 9, pp. 161669-161682 | DOI: 10.1109/ACCESS.2021.3132668 | Scopus / IEEE

Key Finding

BERT-based transformer shows strong contextual understanding across sign sequences. Requires minimum 10,000+ labelled sequences for effective pretraining and significant compute for real-time inference. Contextual attention across adjacent signs improves sentence-level recognition.

What We Found

The 10,000+ sequence requirement is prohibitive for a personal dataset approach. However, their contextual attention finding informed our T5-small integration — T5 provides sentence-level contextual correction that pure LSTM cannot, without requiring a pretrained sign-specific model.

Research Gap	Requires massive labelled dataset (10,000+ sequences) inaccessible for personal use. GPU required for real-time inference. Continuous signing only — no isolated sign recognition or alphabet fingerspelling. No text-to-sign reverse module.
Our Work	T5-small serves as the NLP contextual layer on top of our LSTM predictions, providing sentence-level correction without needing sign-specific pretraining. Personal dataset of 960 sequences achieves 98.96% for isolated ASL recognition.

[13]

MediaPipe: A Framework for Perceiving and Processing Reality*C. Lugaresi et al. (2019)**Proceedings CVPR Workshop on CV for AR/VR* | DOI: N/A (Google Technical Report) | Google Scholar / CVPR**Key Finding**

MediaPipe provides unified real-time perception pipelines (hand tracking 21 landmarks, pose 33 landmarks, face 468 landmarks) on commodity hardware at 30+ fps. Hand landmark model achieves sub-centimeter accuracy on consumer webcams. Normalized coordinates (0-1 relative to image) output by default.

What We Found

The sub-centimeter accuracy claim confirmed our approach. At 30fps on MacBook Air M1, disabling face and pose increased throughput from ~25fps to ~30fps. The normalized coordinate output (0-1 relative to image) explained why raw coordinates are not scale-invariant — a sign at close range and far range produces different absolute values despite identical hand shape.

Research Gap

MediaPipe normalized coordinates are image-relative, not scale-invariant — the same hand shape at different distances produces different coordinate values. No built-in normalization relative to wrist. Holistic pipeline includes unnecessary face and pose data for hand-only recognition tasks.

Our Work

Applied second-layer wrist-origin normalization on top of MediaPipe coordinates. Disabled face and pose processing in inference for speed. The result is a fully scale and position invariant 126-dim feature vector that works at any camera distance.

[14]

Long Short-Term Memory*S. Hochreiter and J. Schmidhuber (1997)**Neural Computation, vol. 9, no. 8, pp. 1735-1780* | DOI: 10.1162/neco.1997.9.8.1735 | Scopus**Key Finding**

LSTM gates (input, forget, output) solve the vanishing gradient problem in RNNs enabling learning of temporal dependencies across 100+ timesteps. The cell state acts as long-term memory that selectively retains or discards information. Particularly effective for sequences where relevant information appears at variable time offsets.

What We Found

The forget gate mechanism explains why 30 frames is appropriate for sign recognition — the gate enables the model to discard irrelevant transition frames while retaining the stable hand configuration of a held sign. Reducing to 20 frames degraded accuracy by ~3%, confirming the 30-frame window.

Research Gap

Standard LSTM does not handle abrupt trajectory changes well — signs involving rapid directional motion (J, Z) are harder to classify than static signs. No spatial attention to focus on relevant hand landmarks. Single-layer LSTM insufficient for complex sign patterns.

Our Work

2-layer stacked LSTM (128 to 64 units) allows first layer to capture low-level temporal patterns and second to capture sign-level boundaries. J remains the only consistently weak class due to its trajectory nature — LSTM-GRU hybrid is planned as targeted fix.

[15]

Video-Based Sign Language Recognition with R(2+1)D and LSTM Networks*J. Huang, J. Chaijaruwanich, and V. Chouvatut (2024)**Proceedings IEEE KST, pp. 214-219* | DOI: 10.1109/KST61284.2024.10499589 | Scopus / IEEE**Key Finding**

R(2+1)D spatiotemporal convolutions combined with LSTM improves continuous SLR. Inference latency is 180ms per prediction on GPU, making it unsuitable for real-time CPU deployment. Landmark-based approaches achieve comparable accuracy with 12x lower latency.

What We Found	The 180ms GPU latency confirmed that video-based approaches cannot meet real-time requirements on CPU hardware. Our LSTM on 126-dim landmarks achieves ~15ms inference on M1 CPU — a 12x improvement. This quantitatively validated landmark extraction as the only viable path for offline CPU deployment.
Research Gap	Requires GPU hardware for real-time inference. Large model size not suitable for edge deployment. No normalization — domain gap problem present. Unidirectional continuous signing only — no alphabet recognition, no reverse text-to-sign module.
Our Work	Landmark-based LSTM achieves ~15ms inference on CPU vs 180ms for video approaches. Fully offline with no GPU requirement. 98.96% accuracy on 32 classes including full alphabet and gesture signs.

[16]

Automatic Translation of Sign Language with Multi-Stream 3D CNN*G. Z. De Castro, R. R. Guerra, and F. G. Guimaraes (2023)**Expert Systems with Applications*, vol. 215, p. 119394 | DOI: 10.1016/j.eswa.2022.119394 | Scopus**Key Finding**

Multi-stream 3D CNN processes RGB video, depth maps, and optical flow. Requires minimum 5,000 sequences per class and dedicated GPU. Optical flow stream contributes 8% accuracy improvement specifically for motion-trajectory signs like J and Z.

What We Found

The optical flow finding for motion signs directly explained why J and Z remain weak in our system — these require trajectory tracking that static frame features cannot capture. Optical flow is too compute-intensive for CPU deployment but their analysis confirmed J/Z difficulty is a feature representation problem, not a data quality issue.

Research Gap

Requires GPU and large dataset (5,000+ sequences per class). Optical flow processing adds 40ms latency. No landmark normalization. Unidirectional — recognition only. No NLP output layer.

Our Work

Personal dataset of 30 sequences per class achieves 98.96% overall. J identified as the only remaining weak class due to trajectory nature — optical flow or LSTM-GRU noted as targeted future fix. Full system operates offline on CPU.

[17]

Sign Pose-Based Transformer for Word-Level Sign Language Recognition*M. Bohacek and M. Hruz (2022)**Proceedings IEEE WACVW*, pp. 182-191 | DOI: 10.1109/WACVW54805.2022.00024 | Scopus / IEEE**Key Finding**

Pose-based transformer using skeletal keypoints is more efficient than image-based transformers. Normalizing keypoints relative to body center significantly improves cross-signer generalization — their cross-signer accuracy improved from 71% to 84% with normalization alone.

What We Found

The 71% to 84% cross-signer improvement from normalization alone gave us quantitative evidence that our wrist-normalization approach would solve our domain gap problem. We adapted their body-center normalization concept to wrist-origin normalization for hand-only features.

Research Gap

Body-center normalization only — does not account for hand scale variation at different camera distances. Transformer architecture requires GPU for real-time operation. Word-level only — no alphabet fingerspelling. No reverse text-to-sign module.

Our Work

Wrist-origin normalization with scale invariance (dividing by max landmark distance) goes beyond their body-center approach — achieving both position AND scale invariance. Model works correctly at any camera distance without retraining.

[18]

Sign Language Recognition using Fusion of Image and Hand Landmarks*R. K. Pathan et al. (2023)**Scientific Reports*, vol. 13 | DOI: 10.1038/s41598-023-43852-x | Scopus**Key Finding**

Fusing image features with hand landmarks improves recognition by 6.3% over landmarks alone for ambiguous signs. Hand landmarks alone achieve 91.2% while fusion achieves 97.5%. The improvement is concentrated on sign pairs with nearly identical hand shapes where orientation cues help disambiguate.

What We Found

The paper's confusion analysis showed that the ambiguous pairs (O/E, B/C, V/W) are exactly the pairs our cosine angle discriminator targets. Their finding that orientation cues are the key disambiguating factor directly motivated measuring angular differences between hand vectors rather than adding image fusion overhead.

**Research
Gap**

Image fusion adds 15-20ms overhead per frame, incompatible with real-time CPU deployment. No normalization applied to landmark features — scale dependency present. No sentence formation or NLP layer. Unidirectional system.

Our Work

Cosine angle discriminator achieves disambiguation in under 1ms with no image overhead. All 6 confused pairs confirmed separable by angle. Wrist-origin normalization applied for scale invariance. T5 gloss layer for sentence formation.

[19]
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Real-Time Hand Gesture Detection and Recognition Using Bag-of-Features and SVM

N. H. Dardas and N. D. Georganas (2011)

IEEE Transactions on Instrumentation and Measurement, vol. 60, no. 11, pp. 3592-3607 | DOI: 10.1109/TIM.2011.2161140 | Scopus / IEEE

Key Finding

Hand region features are sufficient for gesture recognition, SVM achieves 91% on 10 gestures in real-time. Temporal smoothing of predictions over 5 frames eliminates 89% of single-frame noise. Background subtraction and hand isolation are critical preprocessing steps.

What We Found

The 89% noise reduction from 5-frame temporal smoothing gave us confidence that our 10-frame stability buffer was appropriately sized for our 32-class problem. Their hand-region isolation principle also directly validated our approach of discarding pose and face features entirely.

Research Gap

SVM cannot model temporal dynamics of sign sequences — only static hand shapes. 10 gesture classes only, no alphabet coverage. No bidirectional capability. No sentence formation. Background subtraction requires controlled lighting environment.

Our Work

LSTM captures temporal dynamics that SVM cannot. 32 classes including full 26-letter alphabet and 6 gesture signs. 10-frame stability buffer applies temporal smoothing. Works in varied lighting without background subtraction — MediaPipe handles hand detection robustly.

[20]
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Isolated Sign Language Recognition with Multi-Scale Features using LSTM

O. Mercanoglu, A. O. Tur, and H. Keles (2019)

Proceedings IEEE SIU, Sivas, Turkey, pp. 1-4 | DOI: 10.1109/SIU.2019.8806314 | Scopus / IEEE

Key Finding

Multi-scale LSTM with 15, 30, and 60 frame windows tested. 30 frames at 30fps provides optimal balance between capturing complete sign motion and avoiding padding for static signs. Class-weighted training significantly improves minority class recall on imbalanced small datasets. 60 frames showed no accuracy improvement over 30 frames.

What We Found

The 30-frame optimum directly validated our sequence length choice and confirmed we did not need to increase the window. Their class-weighted training finding was confirmed in our experiments — without class weights, gesture signs (6 classes) and letter signs (26 classes) had imbalanced recall despite equal sequence counts.

Research Gap

No normalization — raw coordinate features are scale and position dependent. Unidirectional recognition only. No inference pipeline for real-time deployment discussion. No NLP or sentence formation layer. No text-to-sign reverse module.

Our Work

30-frame sequences with wrist-origin normalized 126-dim features. Balanced class weights via `compute_class_weight`. `EarlyStopping` patience=50 compensates for small dataset. Full real-time inference pipeline at ~15ms on CPU. T5 gloss for sentence formation. Bidirectional system.